

Dharshan Sundarraj

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Portfolio: <https://dharshansundarraj.github.io/Portfolio/>

Professional Summary

Detail-oriented Information Technology graduate and Backend Java Developer with certified foundations from Oracle. Experienced in building offline-resilient web application backends, ACID-compliant REST APIs, and integrated IoT data pipelines. Proficient in Java, Spring Boot, and MySQL with hands-on exposure to responsive user interface integration. Seeking an entry-level Software Engineer or Java Backend Developer role to contribute to scalable systems.

Technical Skills

Languages: Core Java, SQL, JavaScript (ES6+), HTML5, CSS3.

Backend Frameworks & APIs: Spring Boot, RESTful Web Services, JDBC, Maven

Databases: MySQL, Relational Schema Design, Query Optimization, ACID Compliance

Core Competencies: Data Structures & Algorithms (DSA), OOP, Unit Testing (JUnit)

Developer Tools: Git, GitHub, VS Code, Figma, Agile/Scrum, Arduino IDE

Frontend Concepts: Responsive UI Design, Progressive Web Apps (PWA), DOM Manipulation

Technical Projects

TransitPay Ecosystem: Offline-Resilient Transit Ledger | Lead Developer

- **Tech Stack:** Java, Spring Boot, MySQL, Vanilla JavaScript, HTML/CSS, PWA (LocalStorage, Web APIs), html5-qrcode.
- **Offline-First Architecture:** Engineered a Progressive Web App using navigator.onLine and LocalStorage APIs to queue 500+ fare transactions locally during network outages, achieving seamless zero-loss sync upon reconnection.
- **ACID-Compliant Backend:** Architected a Spring Boot REST API with strict @Transactional boundaries and structured exception handling, ensuring exact fare deductions against an immutable MySQL ledger across concurrent sessions.
- **QR Scan Integration:** Integrated html5-qrcode library to enable contactless fare logging in under 2 seconds per scan, reducing manual entry errors to zero.

IoT Smart Authentication System | Lead Full-Stack Developer (Final Year Academic Project)

- **Tech Stack:** Java, MySQL, HTML/CSS/JavaScript, Arduino IDE, ESP32-CAM.
- **Full-Stack Architecture:** Led the complete software development lifecycle for a 4-person team, engineering the end-to-end data pipeline from ESP32-CAM facial recognition capture through a Java server to a live administrative dashboard.
- **Server & Database:** Designed a normalised MySQL schema and built a centralised Java server to securely process, store, and retrieve facial recognition events with sub-1.2-second door-trigger latency.
- **Hardware-Software Integration:** Programmed the ESP32-CAM microcontroller via Arduino IDE to trigger a solenoid lock exclusively upon successful server-side validation, eliminating false-access events.

Digital Ecosystem Redesign (TNSTC & Paper Glass)

- **Tech Stack:** Figma, UI/UX Design, Component Architecture.
- Redesigned the mobile application interface for the Tamil Nadu State Transport Corporation, optimizing user navigation and accessibility.
- Developed "Paper Glass," a comprehensive, responsive UI component library utilizing advanced auto-layout and prototyping logic for future frontend handoffs.

Work Experience

Frontend Development Intern Codveda Technologies

(Virtual) | Dec 2025 – Jan 2026

- Collaborated in virtual environments to translate design mockups into functional, responsive web components for a core system user experience.
- Owned responsibility for implementing UI components with clean, maintainable code and participated in structured code reviews and remote agile workflows.

IoT intern Power Integrated

(Virtual) | May 2025 – Jun 2025

- Developed embedded C programming skills for microcontrollers to interact with temperature, humidity, and light sensors.
- Designed IoT systems for real-time data collection and processing, gaining practical insight into system connectivity.

Education

B.E. in Information Technology - Annamalai University, Cuddalore, Tamil Nadu

CGPA: 7.24 / 10

Relevant coursework: Data Structures, DBMS, Computer Networks, Software Engineering

Certifications

- **UI/UX Wireframing and Prototyping with Figma** – SkillUp / Coursera (Mar 2026)
- **Oracle Java Foundations** – Oracle / Coursera (Mar 2026)
- **Foundations of User Experience (UX) Design** – Google / Coursera (Dec 2025)
- **Introduction to Industry 4.0 and Industrial IoT** – NPTEL (Oct 2025)
- **Privacy and Security in Online Social Media** – NPTEL (Oct 2025)